

Process Flow

Process Name

Customer Onboarding — Branch-Based Account Opening (As-Is)

Objective

To document the current manual, branch-dependent process by which a retail customer opens a new savings account at Zenith National Bank, from initial branch visit through account activation, debit card issuance, and internet banking access.

Actors

- **Customer** — individual seeking to open a new account
- **Customer Service Representative (CSR)** — first point of contact at the branch
- **Branch Officer** — verifies and approves the application
- **Operations Team** — performs backend data entry and card/internet banking requests
- **Core Banking System (CBS)** — legacy system of record for account creation

Step-by-Step Current Process (20 Steps)

1. Customer visits the branch during working hours with original ID and address proof.
2. Customer takes a token and waits in the general queue.
3. CSR greets the customer and understands the requirement (new account opening).
4. CSR manually verifies the customer's original identity and address proof documents (physical KYC check).
5. CSR hands over a blank paper account-opening application form to the customer.
6. Customer manually fills out the paper form (personal details, nominee, account type).
7. Customer attaches self-attested photocopies of ID proof, address proof, and a passport-size photograph.
8. CSR reviews the filled form for completeness and accuracy.
9. **Decision Point:** If the form is incomplete or documents are missing, CSR returns it to the customer for correction (loop back to Step 6).

10. If complete, CSR logs the application manually in a physical register and forwards it to the Branch Officer.
11. Branch Officer manually cross-checks the application against the submitted documents.
12. **Decision Point:** Branch Officer approves or rejects the application based on document authenticity and eligibility (e.g., age, existing account checks).
13. If rejected, the customer is informed verbally or via a follow-up call; process ends.
14. If approved, Branch Officer forwards the physical file to the Operations Team.
15. Operations Team manually keys the customer's data into the Core Banking System (CBS).
16. CBS generates a new account number and stores the customer record.
17. Operations Team submits a manual request to the card division for debit card issuance.
18. Debit card is printed centrally and dispatched to the customer's registered address via courier (typically 7–10 business days).
19. Operations Team separately submits a request to activate internet banking, which is provisioned manually within CBS.
20. Customer receives account details via a physical welcome letter and/or SMS notification once the account is active.

Decision Points

- Application form completeness check (Step 9)
- Branch Officer approval/rejection based on document verification (Step 12)

Manual Activities

- Physical KYC document verification
- Paper form filling by customer
- Manual logging in a physical register
- Manual cross-checking of documents by Branch Officer
- Manual data entry into CBS by Operations Team
- Manual, separate requests for debit card and internet banking activation

System Interactions

- CBS: customer record creation, account number generation

- Card printing/dispatch system: triggered manually, not integrated with CBS in real time
- Internet banking provisioning module: activated manually by Operations Team, not automatic on account creation
- SMS/notification system: triggered only after account is fully active

Pain Points and Bottlenecks

- Customer must visit the branch physically and cannot begin the process online.
- Long, unpredictable wait times due to token-based queueing.
- Form errors discovered only after manual review, requiring resubmission and repeat visits.
- No real-time verification of ID/address proof — relies entirely on human judgment.
- Multiple handoffs (CSR → Branch Officer → Operations Team) with no shared digital workflow, increasing turnaround time and risk of lost files.
- Debit card and internet banking activation are handled as separate, sequential manual requests instead of running in parallel with account creation.
- No visibility for the customer into application status between submission and account activation.
- Physical document storage creates compliance and retrieval risk.

Business Rules

- Customer must be physically present with original documents for KYC — no remote or digital verification permitted.
- Application form must be complete before being forwarded to the Branch Officer.
- Only the Branch Officer can approve or reject a new account application.
- Debit card and internet banking requests cannot be initiated until the account is confirmed created in CBS.
- Customer must be 18 years or older to open an individual account.

Inputs and Outputs

Inputs: Original ID proof, original address proof, passport-size photograph, filled paper application form, nominee details

Outputs: Active bank account with account number, physical welcome letter, debit card (dispatched separately), internet banking credentials, SMS/notification confirming activation

Opportunities for Improvement

- Replace physical KYC with digital eKYC (Aadhaar/PAN-based verification with video KYC).
- Eliminate paper forms in favor of a digital application captured directly into the system.
- Enable parallel processing of debit card and internet banking requests instead of sequential handling.
- Provide real-time application status visibility to the customer.
- Reduce manual handoffs between CSR, Branch Officer, and Operations Team through a shared digital workflow queue.

BPMN-Style Process Flow Description

Pool: Zenith National Bank — Branch Account Opening (As-Is)

Lane: Customer

- Start Event: Customer arrives at branch
- Task: Take token, wait in queue
- Task: Provide original documents for KYC
- Task: Fill paper application form
- Task: Attach photocopies and photograph

Lane: Customer Service Representative (CSR)

- Task: Perform manual KYC verification
- Task: Issue application form
- Task: Review form completeness
- Gateway (Exclusive): Form complete?
 - No → Task: Return form to customer (loop to Customer lane)
 - Yes → Task: Log application, forward to Branch Officer

Lane: Branch Officer

- Task: Cross-check documents against application
- Gateway (Exclusive): Application approved?
 - No → Task: Notify customer of rejection → End Event: Application rejected
 - Yes → Task: Forward file to Operations Team

Lane: Operations Team

- Task: Enter customer data into CBS
- Task: Request debit card issuance
- Task: Request internet banking activation

Lane: Core Banking System (CBS)

- Task: Generate account number and create record
- Task: Provision internet banking access
- End Event: Notification sent to customer, account active

Pain Points Identified by the Business Analyst

1. Physical presence requirement excludes customers who cannot easily visit a branch, contributing to the onboarding delays flagged in the Business Problem.
2. Sequential, non-integrated handoffs between CSR, Branch Officer, and Operations Team introduce delay and risk of file loss with no digital audit trail.
3. Manual KYC and data entry are error-prone and slow compared to automated verification against government databases.
4. Debit card and internet banking are treated as afterthoughts rather than part of a single onboarding transaction, extending time-to-first-use well beyond account creation.
5. Customers have no self-service visibility into where their application stands, driving avoidable branch and call-centre follow-ups.

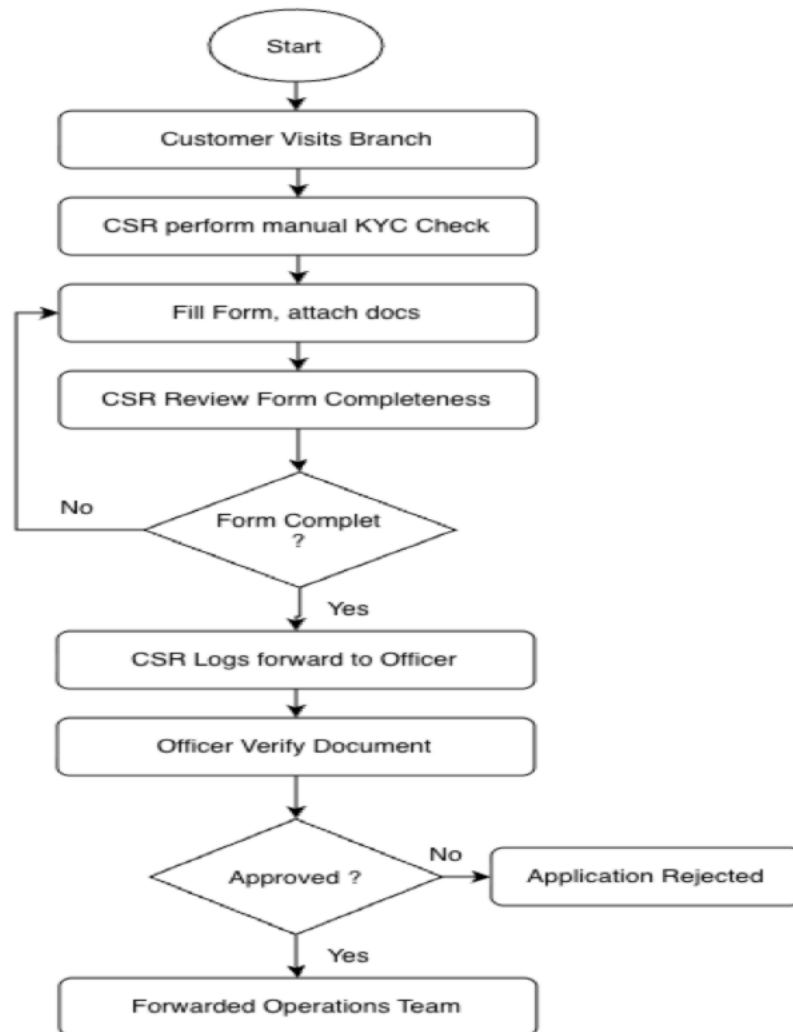
Recommendations for the To-Be Process

- Replace manual KYC with digital eKYC (Aadhaar/PAN validation + video KYC) to remove the branch dependency entirely for standard cases.
- Digitize the application form so customer-entered data flows directly into the platform, removing manual re-keying by Operations Team.
- Introduce a risk-based approval model: auto-approve standard applications, route only flagged/mismatched cases to manual compliance review.

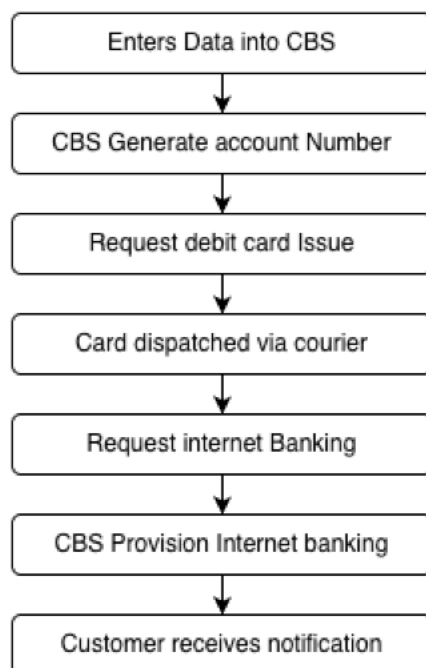
- Trigger debit card issuance and internet banking activation automatically and in parallel upon account creation, rather than as separate sequential requests.
- Provide customers a real-time application status tracker to eliminate status-check calls and visits.

As-Is Process Flow — Part 1: Branch Application & Approval

Part 1: Branch Application & Approval



As-Is Process Flow — Part 2: Backend Processing & Activation



Process Name

Digital Customer Onboarding — Self-Service Account Opening (To-Be)

Business Objective

To enable a retail customer to open a fully functional bank account — including internet banking and a debit card — entirely through the web or mobile app, without visiting a branch, using automated eKYC verification and system-to-system integration to reduce onboarding time from 2-3 days to same-day activation.

Actors

- **Customer** — individual opening a new account via web or mobile app
- **Digital Banking Platform** — the customer-facing application orchestrating the journey
- **eKYC Service** — third-party/government-integrated identity verification service (Aadhaar/PAN, video KYC)
- **Core Banking System (CBS)** — system of record for account creation
- **Notification Service** — sends SMS/email/in-app alerts

- **Bank Administrator** — reviews only flagged/exception cases

Step-by-Step Future Process (18 Steps)

1. Customer downloads the app or visits the web platform and selects "Open New Account."
2. Customer enters mobile number and email address.
3. Platform sends OTP to mobile number and email simultaneously.
4. Customer enters both OTPs to verify contact details.
5. **Decision Point:** Are both OTPs valid? If invalid, customer is prompted to re-enter or request a resend (loop to Step 3).
6. Customer enters PAN and Aadhaar number.
7. Platform calls the eKYC Service to validate identity against UIDAI/NSDL in real time.
8. Customer uploads a photo ID and completes a live video KYC session within the app.
9. eKYC Service cross-checks the video KYC capture against the submitted ID document (face match, liveness check).
10. **Decision Point:** Does eKYC pass validation? If a mismatch or low-confidence match is detected, the application is routed to the Bank Administrator for manual review; customer is notified their application is under review (exception path).
11. If eKYC passes automatically, the platform proceeds to digital account opening — customer selects account type and reviews terms.
12. Customer digitally accepts terms and conditions and submits the application.
13. Platform sends the validated application data to the Core Banking System via API.
14. CBS automatically creates the account and generates an account number in real time.
15. Platform triggers instant Internet Banking activation upon account creation confirmation from CBS.
16. Customer sets up login credentials and registers for Multi-Factor Authentication (OTP and/or biometric).
17. Platform submits a parallel automated request for debit card issuance (physical + optional instant virtual card).
18. Notification Service sends a real-time welcome email and SMS confirming the account number, internet banking activation, and card dispatch status.

Decision Points

- OTP validity check for mobile and email (Step 5)

- eKYC pass/fail validation, determining automated approval vs. manual review routing (Step 10)

Automated System Activities

- OTP generation and validation
- Real-time PAN/Aadhaar validation against government databases
- Face match and liveness detection during video KYC
- Automatic account number generation in CBS
- Automatic Internet Banking provisioning
- Automatic debit card issuance request
- Automated real-time notifications at each key milestone

System Integrations

- Digital Banking Platform ↔ eKYC Service (identity verification API)
- Digital Banking Platform ↔ Core Banking System (account creation API)
- Digital Banking Platform ↔ Notification Service (SMS/email/push gateway)
- Core Banking System ↔ Card Management System (debit card issuance trigger)
- Digital Banking Platform ↔ MFA Provider (OTP/biometric authentication service)

Business Rules

- Mobile number and email must each be verified via OTP before proceeding to eKYC.
- Customer must be 18 years or older; the system shall auto-reject underage applicants at the PAN/Aadhaar validation step.
- Video KYC must achieve a minimum face-match confidence threshold; below this, the application routes to manual review, never auto-rejects.
- Internet Banking shall only activate after CBS confirms successful account creation.
- MFA registration is mandatory before first login is permitted.
- Debit card issuance request shall trigger automatically and in parallel with Internet Banking activation, not sequentially.

Inputs and Outputs

Inputs: Mobile number, email address, PAN, Aadhaar number, live video KYC capture, digitally accepted terms and conditions

Outputs: Active bank account with account number, activated Internet Banking credentials, MFA registration, debit card dispatch request confirmation, real-time welcome email/SMS

Benefits Over the As-Is Process

As-Is	To-Be
2-3 working days for account activation	Same-day, often within minutes
Mandatory branch visit	Fully remote via web/mobile
Manual KYC verification	Automated eKYC with video verification
Paper application form	Fully digital data capture
Sequential debit card and IB activation	Parallel, automatic activation
Multiple manual handoffs (CSR → Officer → Ops)	Single automated workflow with exception-only human review
No visibility into application status	Real-time in-app status and notifications
Single-factor login	Mandatory MFA from day one

Exception Scenarios

- **OTP not received:** Customer can request a resend after a cooldown period; repeated failures direct the customer to alternate verification (registered branch or support).
- **eKYC face-match failure:** Application routed to Bank Administrator queue for manual review; customer notified of delay, not rejection.
- **PAN/Aadhaar validation service downtime:** Application is held in a pending state with customer notification; automatic retry attempted before manual fallback.
- **CBS account creation failure:** Platform retries automatically; if unresolved, the case is escalated to the Bank Administrator with the customer notified of the delay.
- **Underage applicant:** System auto-rejects at PAN/Aadhaar validation with a clear on-screen message; no manual review needed

BPMN-Style Process Flow Description

Pool: Zenith National Bank — Digital Customer Onboarding (To-Be)

Lane: Customer

- Start Event: Customer opens app/web platform
- Task: Enter mobile number and email
- Task: Enter OTPs received
- Task: Enter PAN and Aadhaar
- Task: Complete video KYC
- Task: Select account type, accept terms
- Task: Set up MFA credentials

Lane: Digital Banking Platform

- Task: Send OTP to mobile and email
- Gateway (Exclusive): OTPs valid?
 - No → Task: Prompt resend (loop to Customer lane)
 - Yes → Task: Call eKYC Service
- Task: Submit application data to CBS
- Task: Trigger Internet Banking activation
- Task: Trigger parallel debit card request

Lane: eKYC Service

- Task: Validate PAN/Aadhaar against government database
- Task: Perform face-match and liveness check on video KYC
- Gateway (Exclusive): eKYC passed?
 - No → Task: Route to Bank Administrator (manual review)
 - Yes → Task: Return validation success to platform

Lane: Core Banking System

- Task: Create account record
- Task: Generate account number
- Task: Confirm creation to platform

Lane: Notification Service

- Task: Send OTP messages
- Task: Send welcome email/SMS with account and activation details
- End Event: Customer onboarded and notified

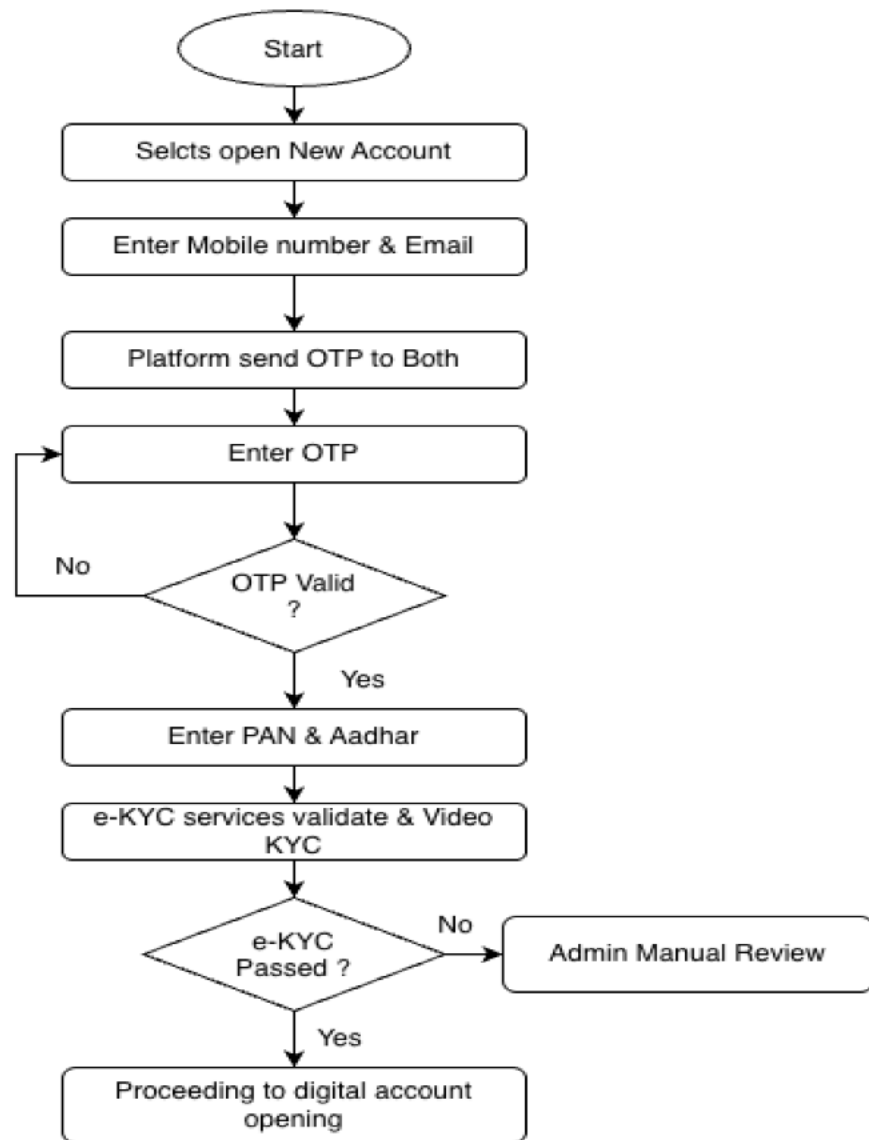
Lane: Bank Administrator

- Task: Review flagged eKYC exceptions
- Gateway (Exclusive): Manual approval granted?
 - o No → End Event: Application rejected, customer notified
 - o Yes → Task: Resume automated flow (rejoin main path)

Comparison: As-Is vs. To-Be

Aspect	As-Is	To-Be
Channel	Branch only	Web and mobile app
Identity verification	Manual, physical document check	Automated eKYC with video verification
Application format	Paper form	Fully digital
Turnaround time	2-3 working days	Same-day, often minutes
Human involvement	CSR, Branch Officer, Operations Team at every step	System-automated; human involved only for flagged exceptions
Card & Internet Banking activation	Sequential, separately requested	Parallel, automatic
Authentication	Single-factor login	Mandatory MFA
Status visibility	None until completion	Real-time in-app and notification updates one
Compliance/audit trail	Manual register, physical files	Fully digital, system-logged

To-Be Process Flow — Part 1: Verification & eKYC



To-Be Process Flow — Part 2: Account Creation & Activation

